

EMERGENCY RESPONSE SYSTEM



CSE 221: Data Structure

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April 29, 2026

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Abstract

This project presents a **terminal-based Smart Emergency Response Management System** developed using **C++**, focusing on the application of core data structures. The system simulates how emergency services manage incoming cases, prioritize them based on severity, and assign ambulances efficiently.

Unlike traditional systems that rely on databases, this implementation uses multiple data structures such as **Priority Queue, Stack, Vector, Map, and Graph**. Each structure is chosen for its efficiency in handling specific operations.

The project demonstrates how combining different data structures can result in an optimized and realistic emergency management system.

Chapter 1: Introduction

1.1 Problem Statement

Managing emergency situations efficiently is a critical challenge. Without proper prioritization and resource allocation, response time increases, which can lead to serious consequences. A system is needed to handle emergencies based on severity and assign available ambulances effectively.

1.2 Objectives

- To develop a terminal-based emergency system using C++
- To implement multiple data structures in a real-world scenario
- To prioritize emergencies based on severity
- To manage ambulance allocation efficiently
- To implement undo functionality

1.3 Flow of the Project

The system starts with a menu-driven interface. The user inputs emergency details, which are stored and processed using appropriate data structures. Based on severity and location, the system assigns ambulances and updates status dynamically.

Chapter 2: Background

2.1 Existing System Analysis

Most modern emergency systems rely on complex databases and real-time tracking systems. While efficient, they do not clearly demonstrate the internal working of data structures. This project focuses on implementing core data structures directly in a simplified environment.

2.2 Supporting Concepts

This project is based on fundamental data structure concepts including:

- Priority-based processing
- Stack operations for undo
- Dynamic arrays
- Graph-based distance calculation

Chapter 3: System Analysis and Design

3.1 Core Data Structures

Priority Queue (Max Heap)

Used to store emergency requests and always process the highest severity first.

Time Complexity: $O(\log n)$

Stack

Used for undo operations (Last-In-First-Out behavior).

Time Complexity: $O(1)$

Vector

Stores ambulance data and completed emergencies dynamically.

Time Complexity: $O(1)$ access

Map

Used for mapping area names and quick lookup.

Time Complexity: $O(\log n)$

Graph (Adjacency Matrix)

Represents distances between locations to find the nearest ambulance.

3.2 Data Flow

- Emergency input $\rightarrow$ Priority Queue
- Assignment $\rightarrow$ Graph + Vector
- Undo $\rightarrow$ Stack
- Completed $\rightarrow$ Vector

3.3 Data Structure Summary

Feature	Data Structure	Reason
Emergency Priority	Priority Queue	Highest severity first
Undo	Stack	LIFO behavior
Ambulance Storage	Vector	Dynamic and fast
Area Mapping	Map	Fast lookup
Distance	Graph	Nearest selection

Chapter 4: Implementation

4.1 Interface Design

The system uses a console-based menu-driven interface similar to the sample report .

4.2 Functional Modules

- `addEmergency()` → Adds emergency case
- `assignAmbulance()` → Assigns nearest ambulance
- `showPending()` → Displays pending cases
- `showCompleted()` → Displays handled cases
- `undo()` → Reverts last operation
- `returnAmbulance()` → Marks ambulance free

4.3 Backend Logic

All data is stored in memory using data structures. No external database or file system is used. The system ensures efficient operations using optimized algorithms.

Chapter 5: User Manual

5.1 System Requirements

Hardware:

- Any standard computer

Software:

- Operating System: Windows/Linux/macOS
- Compiler: G++
- Environment: Terminal / Command Prompt

5.2 User Operations

Main Menu:

User selects options from menu

```
==== EMERGENCY SYSTEM ====
1. Add Emergency
2. Assign Ambulance
3. Show All Ambulances
4. Hospital Status
5. Pending Emergencies
6. Completed Emergencies
7. System Stats
8. Undo
9. Return Ambulance
10. Exit
Choice:
```

Add Emergency:

User inputs severity and location

```
Choice: 1
Enter Area (Mirpur/Dhanmondi/Gulshan/Uttara): mirpur
Enter Severity (1-5): 5
Emergency Added! ID: 1
```

Assign Ambulance:

System automatically assigns based on priority

```
Choice: 2
```

```
Dispatch Successful  
Ambulance ID: 1  
Hospital: DMC  
Distance: 0 km
```

Undo:

Reverts last action

```
Choice: 8  
Undo successful
```

Return Ambulance:

Marks ambulance available again

```
Choice: 9  
Enter Ambulance ID: 1  
Ambulance 1 returned to MIRPUR
```

Chapter 6: Conclusion

6.1 Conclusion

This project successfully demonstrates how multiple data structures can be integrated to build an efficient emergency response system. Each structure contributes to optimized performance and realistic behavior.

6.2 Limitations

- No persistent storage
- Static distance system
- No real-time tracking
- Console-based interface

6.3 Future Work

- Implement Dijkstra algorithm
- Add file/database storage
- Develop graphical interface
- Expand real-world mapping

References

- Bjarne Stroustrup, *The C++ Programming Language*
- Mark Allen Weiss, *Data Structures and Algorithms*
- C++ Standard Library Documentation